exercise 7.

Problem 1.

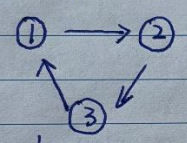
1. Great few examples of (directed) graphs and their transposed graphs

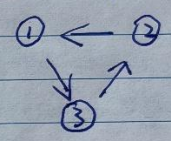
Def: The transpose of a directed graph $G = (V, E)$ is the graph $G^T = (V, E^T)$, where $E^T = \{(v, u) : (u, v) \text{ in } E\}$. Every edge direction is reversed.

ex 1: $G1$ has $V = \{1, 2, 3\}$, $E = \{(1,2), (2,3), (3,1)\}$.

Transpose $G1^T$ has $E^T = \{(2,1), (3,2), (1,3)\}$.

$G1 : E = \{(1,2), (2,3), (3,1)\}$ $\quad$ $G1^T = E = \{(2,1), (3,2), (1,3)\}$

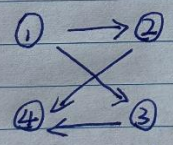


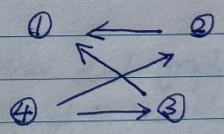


ex2 $G2$ has $V = \{1, 2, 3, 4\}$, $E = \{(1,2), (1,3), (2,4), (3,4)\}$.

Transpose $G2^T$ has $E^T = \{(2,1), (3,1), (4,2), (4,3)\}$.

$G2 : E = \{(1,2), (1,3), (2,4), (3,4)\}$ $\quad$ $G2^T : E = \{(2,1), (3,1), (4,2), (4,3)\}$





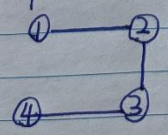
2. (undirected) graphs

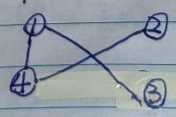
Def: The complement of an undirected graph $G(V, E)$ is the graph $\bar{G} = (V, \bar{E})$, where $\bar{E}$ contains exactly those edges NOT in E (for distinct vertices).

ex 1: $G3$ is a path $P4$, $V = \{1, 2, 3, 4\}$, $E = \{(1,2), (2,3), (3,4)\}$.

Complement has $\bar{E} = \{(1,3), (1,4), (2,4)\}$

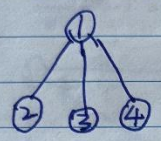
$G3$ (path) $: E = \{(1,2), (2,3), (3,4)\}$ $\quad$ Complement $: E = \{(1,3), (1,4), (2,4)\}$



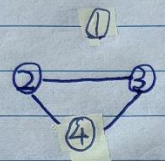


ex2. G_4 is a star with center 1. $E = \{(1,2), (1,3), (1,4)\}$.
Complement has $\bar{E} = \{(2,3), (2,4), (3,4)\}$ – a triangle on $\{2,3,4\}$

G_4(star) = $E = \{(1,2), (1,3), (1,4)\}$



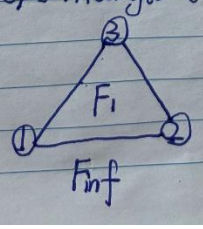
Complement : $E = \{(2,3), (2,4), (3,4)\}$



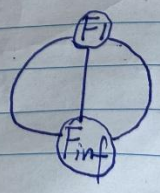
3. what happens if the original is dense for the inverse?
If the original graph G is dense ($|E|$ close to $\frac{n(n-1)}{2}$), then its complement $\bar{G}$ will be sparse. In the extreme, the complement of a complete graph K_n is the empty graph (no edges). This is because $|E(G)| + |E(\bar{G})| = n(n-1)/2$ always holds. Conversely, a sparse graph has a dense complement.

4. examples of (undirected) graphs and their dual graphs.
Def: Given a planar graph G embedded in the plane, its dual graph G^* is constructed by: ① placing one vertex in each face of G (including the outer face).
② for each edge e of G, connecting the dual vertices of the two faces on either side of e.
ex1: A triangle (3 vertices, 3 edges) has 2 faces (1 inner face F1, 1 outer face Finf). The dual G^* has 2 vertices connected by 3 parallel edges.

G: Triangle (2 faces)

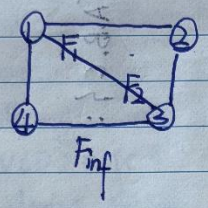


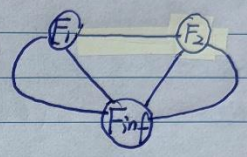
G^* (Dual): 2 nodes, 3 edges.



ex2: A square with one diagonal (4 vertices, 5 edges) has 3 faces (F_1, F_2, F_{inf}) The dual G^* has 3 vertices with 5 edges.

G: Square + diagonal (3 faces) — G^* (Dual): 3 nodes, 5 edges





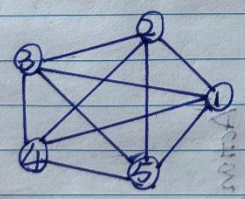
5. The dual construction requires a planar embedding so that well-defined faces exist. A non-planar graph cannot be draw in the plane without edge crossings, so it has no well-defined faces, and no dual can be constructed.

Non-planar example – K5 (complete graph on 5 vertices):

K5 has 5 vertices and 10 edges. By Kuratowski's theorem, a graph is planar iff it contains no subdivision of K5 or K3,3. K5 trivially contains itself. Also, for planar graphs $|E| \leq 3|V| - 6$, but $10 > 3(5) - 6 = 9$.

Therefore K5 is non-planar and has no dual.

K5: 5 vertices, 10 edges.



problem 2.

1. Write the initial call to the algorithm.

Bronkerbosch ($R = \{\}$, $P = \{A, B, C, D\}$, $X = \{\}$).

R = Current clique (empty). P = candidate vertices.

X = already-processed vertices (for avoiding duplicates).

2. Trace the first two recursive calls that lead to reporting a maximal clique. Show the values of (R, P, X) at each step.

Adjacency : $N(A) = \{B, C\}$, $N(B) = \{A, C\}$, $N(C) = \{A, B, D\}$, $N(D) = \{C\}$.

Finding clique $\{A, B, C\}$:

Step	Call	R	P	X	Action/Result
1	Top-level, pick $v=A$	$\{\}$	$\{A, B, C, D\}$	$\{\}$	Recurse with $R \cup \{A\}$.
1.1	Pick $v=B$	$\{A\}$	$P \cap N(A) = \{B, C\}$	$X \cap N(A) = \{\}$	Recurse with $R \cup \{B\}$
1.1.1	Pick $v=C$	$\{A, B\}$	$P \cap N(B) = \{C\}$	$X \cap N(B) = \{\}$	Recurse with $R \cup \{C\}$
1.1.1.1	Leaf	$\{A, B, C\}$	$P \cap N(C) = \{\}$	$X \cap N(C) = \{\}$	$P = \emptyset$, $X = \emptyset \rightarrow$ REPORT $\{A, B, C\}$

Finding clique $\{C, D\}$:

Step	Call	R	P	X	Action/Result
2	Top-level, pick $v=B$	$\{\}$	$\{B, C, D\}$	$\{A\}$	$P \cap N(B) = \{C\}$, $X \cap N(B) = \{A\} \neq \emptyset$
2.1	Pick $v=C$	$\{B\}$	$\{C\}$	$\{A\}$	Recurse
2.1.1	Leaf	$\{B, C\}$	$P \cap N(C) = \{\}$	$X \cap N(C) = \{A\} \neq \emptyset$	$X \neq \emptyset \rightarrow$ prune (not maximal)
3	Top-level, pick $v=C$	$\{\}$	$\{C, D\}$	$\{A, B\}$	$P \cap N(C) = \{D\}$, $X \cap N(C) = \{A, B\}$.
3.1	Pick $v=D$	$\{C\}$	$\{D\}$	$\{\}$	Recurse with $R \cup \{D\}$
3.1.1	Leaf	$\{C, D\}$	$P \cap N(D) = \{\}$	$X \cap X(D) = \{\}$	$P = \emptyset$, $X = \emptyset \rightarrow$ REPORT $\{C, D\}$
4	Top-level, pick $v=D$	$\{\}$	$\{D\}$	$\{A, B, C\}$	$P \cap N(D) = \{\}$, $X \cap N(D) = \{C\} \neq \emptyset \rightarrow$ prune

So. Starting from the top level, picking $v=A$ leads us down to $\{A, B, C\}$ which is reported as maximal ($P=X=\emptyset$). After backtracking and processing B (pruned because $X \cap N(B)$ is non-empty at the leaf), we pick $v=C$ at the top level. This leads to $\{C, D\}$ being reported as maximal. Finally, picking $v=D$ is pruned immediately.

3. List all maximal cliques of G. Which one(s) are maximum?

The Bron-kerbosch algorithm finds exactly two maximal cliques:

	Maximal Clique	Size	Maximum.
1	$\{A, B, C\}$	3	Yes.
2.	$\{C, D\}$	2	No.

The maximum clique is $\{A, B, C\}$, with size 3. It has the largest number of vertices among all maximal cliques.

$\{C, D\}$ is also a maximal clique (it cannot be extended further), but its size is only 2, so it is not the maximum clique.